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Assignment -2

* Section - A

Ans 1 CPU registers are small, high - speed storage location inside the processors.

Importance

- Store intermmediate data and instructions
- Faster than cache and memory
- Improve execution speed of programe.

Ans 2 Program counter (PC) is a regester that holds the address of the Next instructions

Function

- Points to next instruction to be executed.
- Automatically updates after each intruction
- controls instruction sequence.

Ans 3 Instruction format defines the structure of an instruction in Binary forms

Key fields

- Oppose (operation)
- Operand
- Addressing mode

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Ans 5 Addressing mode specify how to locate operands

Purpose
- Provide flexibility in accessing data.
- Reduce instruction count.
- Improve programe efficiency

Ans 6 ISA is the interface between hardware and software defining supported instruction

Includes
- Instruction formats
- Data types
- Registers and addressing modes

Ans 7 Types of instruction in a processor are
- Arithmetic (APP, SUP)
- Logical (ADP, OR)
- Data Transfer (LOAD, STORE)
- control (JUMP, BRANCH)

Ans 8 Explanation

- Opcode is interpreted
- Required operation is identified
- Control signals are generated
- 'Operands are prepared for execution

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Ans 9 Basic Instruction cycle are

- Fetch instruction from memory
- Decide instruction
- Execute operation
- Store result.

Ans 10 Characteristics of RISE Processor are :-

- Simple and fixed-length instruction.
- few addressing modes
- Faster execution (pipline - friendly)
- Load / store architecture

Ans 11 Characteristics of CUC Processor are :-

- Complex and variable - length instruction.
- Many addressing modes
- Fewer instruction per program
- Slower But powerful instruction

Ans 12 RISC - V

- Open - source ISA
- Simple, fixed length instruction
- Easier to design and extend

X80

- Proprietary (Inter / AMD)
- Complex, varaiable - length instructions

* Section - B

Ans 1

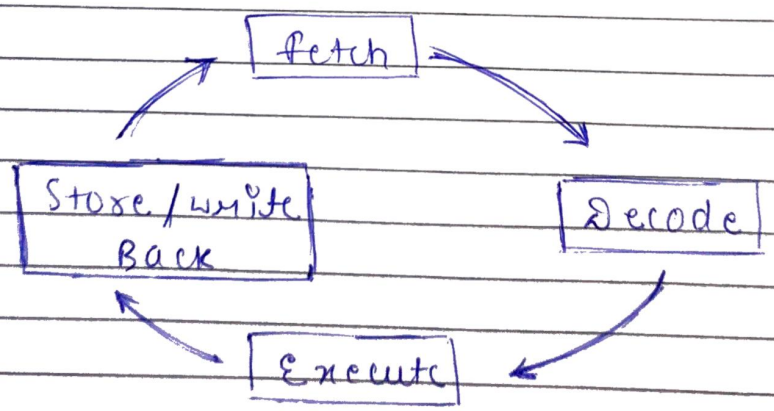
Program counter
(PC)
↓
Instruction Reg
(IA
↓
Memory Address
(MAA)
↓
Accumulator (ACC)

Memory Data
Reg
(MDA)

Explanation

- PC holds next instruction address, IR store current instruction
- MAR & MDR handle memory access, Acc stores intermediate Results

Ans 2



CPU fetches instruction, decodes it, execute it and write Back

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Ans 3 | Opcode | Operand |

Explanation
- Opcode specifies the operations
- Operands specify data or memory location

features	RISE	CISE
Instruction	Simple	Complex
Length	fixed	Variable
speed	fast	slower
Memory	More usage	Less usage

Explanation
- RISE focuser on simplicity and speed.
- CISE focuses on powerful

* Section-C

Ans1 Impact on performance are
- Memory access slower than register access
- Increases CPI due to memory latency
- Overall execution time increase → performace decrease

Ans2 The should choose RISE architecture Because
- Sample and fixed - length instruction
- Easier to pipeline efficiently
- Requires simpler hardware design

Ans 3 CPU executes this instruction as :-

Step ① fetch
Instruction is fetched from memory into IR.

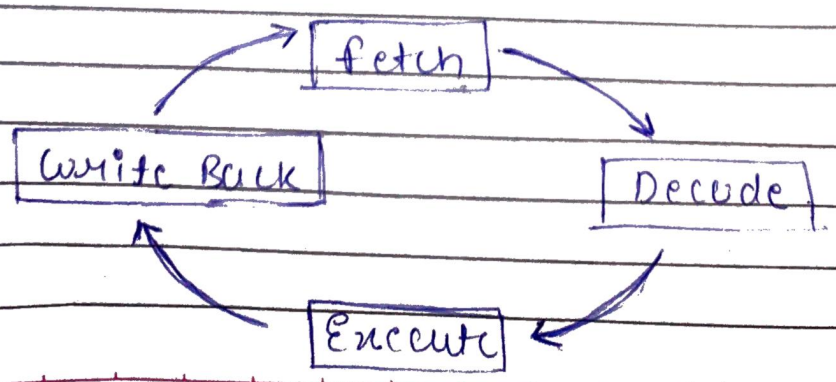
Step ② Decode
Opcode is decoded and register are identified

Step ③ Execute
- Data is read from register
- ALU performs operation
- Result is stored Back in register

Ans 4 X86 is widely used despite complexity Because :-
- Backward compatibility with older software
- Range ecosystem (OS, application, industry support)
- Modern CPUs internally optimited
- Despite complexity, compatibility and ecosystem keep x86 dominant

* Section - D

Ans 1



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Explanation of fetch stage

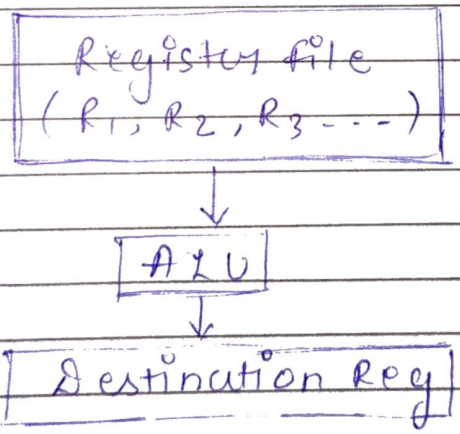
- i.e holds address of nxt instructions
- Address sent to memory via MAR
- Instruction fetched into MDR and loaded into IR.
- PC is imetcmented to next instruction

funct 7 RS 2 rs 1 funct 3 rd opcode

Explanation

- Opcode - specifies operation type
- rd - Destination register
- rs1, rs2 - source register
- funct 3, and funct 7 - Define exact operation
- fixed length (32 - Bit) format simplies doing

Ans 3



Explanation

- Data is read from source registers
- Sent to ALU for operation
- Result is written Back to destination
- Enable fast competation without memory access